

Question number	Answer	Additional guidance	Mark
7(a)	A description that includes the following points: • (for) as blood cholesterol levels increase (up to a point) the number of people developing heart disease increases / at high cholesterol levels more people have heart disease (1) • (against) people with low levels of cholesterol are developing heart disease (1)	ACCEPT at (very) high cholesterol levels there are {only people with heart disease / no people who do not have heart disease} ACCEPT {as cholesterol levels increase / at high levels of cholesterol} there are people without heart disease not enough information on the design of the study	(2)

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7(b)	A description that includes the following points: • (LDL increases so does the incidence of heart disease) the height of the bars increase at all HDL concentrations_(1) • (HDL increases the incidence of heart disease decreases) the bars get smaller at all LDL concentrations (1) • (higher the ratio of HDL to LDL the lower the incidence of heart disease) the bars are highest in the back left corner and get smaller towards the front right corner (1)	ACCEPT CVD is high when LDL high and HDL low <u>and</u> CVD is low when HDL is high and LDL is low	(3)

Question number	Answer	Additional guidance	Mark
7(c)(i)	An explanation that includes three of the following points: • because the LDL-R will {bind / fit} to the LDL (1) • and hold it on the {membrane / cell} (1) • membrane forms a vesicle (around the LDL) (1) • due to the fluidity of the membrane (1)	ACCEPT they are complementary ACCEPT description e.g. membrane surrounds the LDL ACCEPT because phospholipids can move (within membrane)	(3)

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7(c)(ii)	An explanation that includes two of the following points: • because the change in pH will change the ionisation of the R groups (1) • so the {binding site / receptor / protein / LDL} will {change shape / denatures} (1) • therefore bonds holding LDL and LDL-R break (1)	ACCEPT charges of R groups ACCEPT active site {H bonds / ionic bonds} {break / form} within {LDL / LDL-R}	(2)

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7(c)(iii)	An explanation that includes three of the following points: • because mutation results in a change the (tertiary) structure of the {receptor / protein / LDL-R} (1) • therefore the LDL-R will no longer {fit / bind} to LDL / endocytosis will not occur (1) • so less LDL is taken into the {cells / liver} / so LDL remains in the blood (1) • and therefore there will be more {cholesterol / LDL} (in the blood) to {bind to any damaged endothelial cells / build up in the walls of the arteries} (1)	ACCEPT {fewer / no} receptors different protein formed active site ACCEPT no longer complementary ACCEPT LDL increases in blood ACCEPT to form {plaque / atheroma} (in the walls of the arteries)	(3)